INFINITE TUTORIAL - OFFICIAL EXAMINATION PAPER

KARNATAKA STATE BOARD - CLASS 9 MATHEMATICS (NUMBER SYSTEMS)

Max Marks: 50 | Duration: 90 Minutes | Subject Code: MAT9

General Instructions:
1. All questions are compulsory. Internal choices are provided in Section C and D.
2. Section A has 4 MCQs (1 Mark each).
3. Section B has 3 Short Answer Questions (2 Marks each).
4. Section C has 4 Long Answer Questions (3 Marks each).
5. Section D has 4 Analytical Questions (7 Marks each).

SECTION A - MULTIPLE CHOICE QUESTIONS (1 Mark Each)
Q1. Which of the following is an irrational number? [A] sqrt(4) [B] sqrt(9) [C] sqrt(7) [D] 0.25
Q2. The decimal representation of a rational number cannot be: Non-terminating non-repeating.
Q3. Find the value of (64)^(1/2). Ans: 8.
Q4. The sum of a rational and an irrational number is always: Irrational.

SECTION B - SHORT ANSWER QUESTIONS (2 Marks Each)
Q5. Express 0.3333... in the form p/q, where p and q are integers and q != 0.
Q6. Find two rational numbers between 3/5 and 4/5.
Q7. Rationalise the denominator of 1 / (sqrt(7) - sqrt(6)).

SECTION C - LONG ANSWER QUESTIONS (3 Marks Each)
Q8. Represent sqrt(5) on the number line using Pythagoras theorem.
Q9. Simplify: (3 + sqrt(3)) * (2 + sqrt(2)) and expand fully.
Q10. If a = 2 + sqrt(3), find the value of a + 1/a.
Q11. Simplify: (256)^(0.16) * (256)^(0.09).

SECTION D - ANALYTICAL PROBLEMS (7 Marks Each)
Q12. If (sqrt(3) - 1)/(sqrt(3) + 1) = a + b*sqrt(3), determine the rational values of a and b.
Q13. Prove that sqrt(2) is an irrational number using method of contradiction.

*** End of Question Paper - Certified by Infinite Tutorial Academic Directorate ***